

Credit Risk Prediction Using Machine Learning

Python

Scikit-Learn

Status

License

Project Overview

Credit Risk Prediction is a Machine Learning project developed to predict whether a loan application should be approved or rejected based on applicant financial and demographic information.

The primary objective is to assist financial institutions in making data-driven lending decisions while minimizing credit risk and potential financial losses.

Business Problem

Banks receive thousands of loan applications every day. Approving loans to high-risk customers can result in loan defaults and financial losses.

This project aims to build a predictive model that helps identify eligible loan applicants based on their financial profile and credit history.

Dataset Information

Dataset: Loan Prediction Dataset

Total Records: 614

Target Variable:

- Loan_Status
- Y = Approved
- N = Rejected

Features

- Gender

- Married
 - Dependents
 - Education
 - Self_Employed
 - ApplicantIncome
 - CoapplicantIncome
 - LoanAmount
 - Loan_Amount_Term
 - Credit_History
 - Property_Area
-

Technologies Used

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Scikit-Learn
 - XGBoost
 - Google Colab
-

Project Workflow

1. Data Cleaning

- Handled missing values using Mode and Median.
- Checked dataset consistency.

2. Exploratory Data Analysis (EDA)

- Loan Approval Distribution
- Credit History Analysis
- Education Analysis
- Property Area Analysis
- Income Analysis

3. Feature Engineering

Created:

$\text{Total_Income} = \text{ApplicantIncome} + \text{CoapplicantIncome}$

4. Data Preprocessing

- Label Encoding
- Feature Transformation
- Train-Test Split

5. Model Development

Implemented multiple Machine Learning algorithms:

- Logistic Regression
- Random Forest
- XGBoost
- Gaussian Naive Bayes
- Tuned Random Forest (GridSearchCV)

6. Model Evaluation

Evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Model Performance

Model	Accuracy
Logistic Regression	78.86%
Random Forest	77.24%
XGBoost	77.24%
Gaussian Naive Bayes	78.86%
Tuned Random Forest	78.86%

Final Selected Model

Logistic Regression

Training Accuracy: **81.67%**

Testing Accuracy: **78.86%**

Difference: **2.81%**

Model Status

✓ Good Fit

The model generalizes well to unseen data and does not show significant overfitting.

Feature Importance Analysis

Top Factors Influencing Loan Approval

Feature	Importance
Credit_History	25.8%
Total_Income	15.9%
LoanAmount	14.7%
ApplicantIncome	14.5%
CoapplicantIncome	8.7%

Key Business Insights

- Credit History is the strongest predictor of loan approval.
 - Higher applicant income increases approval probability.
 - Total household income plays a major role in lending decisions.
 - Loan amount significantly influences risk assessment.
 - Demographic factors have relatively low impact compared to financial factors.
-

Results

Confusion Matrix

TP = 79

TN = 18

FP = 25

FN = 1

Key Achievement

Successfully developed a machine learning model capable of supporting loan approval decisions with approximately 79% prediction accuracy.

Project Structure

```
Credit-Risk-Prediction-ML/  
|  
├── data/  
│   ├── train.csv  
│   └── test.csv  
├── notebooks/  
│   └── Credit_Risk_Prediction.ipynb  
├── models/  
│   └── loan_model.pkl  
├── images/  
├── README.md  
├── requirements.txt  
└── .gitignore
```

Future Enhancements

- Power BI Dashboard Integration
 - Streamlit Web Application
 - SHAP Explainability
 - ROC-AUC Evaluation
 - Cloud Deployment
 - Real-Time Loan Prediction API
-

Resume Highlights

- Built an end-to-end Machine Learning solution for loan approval prediction.
 - Performed data cleaning, feature engineering, and exploratory data analysis.
 - Compared multiple classification algorithms and optimized model performance using GridSearchCV.
 - Achieved 78.86% prediction accuracy with good model generalization.
 - Generated actionable business insights for credit risk assessment.
-

Author

Chavagani Nagendrababu

Data Analyst | SQL | Python | Power BI | Machine Learning

GitHub: <https://github.com/chavaganinagendrababu-ctrl>

Portfolio: <https://chavaganinagendrababu-ctrl.github.io/Portfolio/>